Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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Methods

Inclusion Criteria

Subjects must satisfy the following criteria to be enrolled in the study:

- Men or women aged ≥18 years with non-valvular atrial fibrillation
- Atrial fibrillation must be documented by ECG evidence (e.g., 12-lead ECG, rhythm strip, Holter, pacemaker interrogation) within 30 days before randomization. In addition, subjects must have medical evidence of atrial fibrillation within 1 year before and at least one day before the qualifying ECG evidence. This could be obtained from a notation in the subject's record (e.g., medical chart, hospital discharge summary).
- Subjects with newly diagnosed atrial fibrillation are eligible provided that:
  - there is evidence that the atrial fibrillation is non-valvular
  - cardioversion is not planned
  - there is ECG evidence on 2 occasions 24 hours apart demonstrating atrial fibrillation
- History of prior ischemic stroke, TIA or non-CNS systemic embolism believed to be cardioembolic in origin or has 2 or more of the following risk factors:
  - Heart failure and/or left ventricular ejection fraction ≤35%
  - Hypertension (defined as use of antihypertensive medications within 6 months before the screening visit or persistent systolic blood pressure above 140 mmHg or diastolic blood pressure above 90 mmHg)
  - Age ≥75 years
  - Diabetes mellitus (defined as a history of type 1 or type 2 diabetes mellitus or use of antidiabetic medications within 6 months before screening visit)
- Female subjects must be postmenopausal (for at least 2 years), surgically sterile, abstinent, or, if sexually active, be practicing an effective method of birth control (e.g., prescription oral contraceptives, contraceptive injections, intrauterine device, double-barrier method, contraceptive patch, male partner sterilization) before entry and throughout the study; and, for those of childbearing potential, have a negative serum β-hCG pregnancy test at screening.

- Subjects must have signed an informed consent document indicating that they understand the purpose of and procedures required for the study and are willing to participate in the study
- In order to participate in the optional pharmacogenomic component, subjects must have signed the informed consent for DNA research document indicating willingness to participate in the pharmacogenomics component of the study (where local regulations permit)

Exclusion Criteria

Potential subjects who meet any of the following criteria will be excluded from participating in the study:

Cardiac-Related Conditions

- Hemodynamically significant mitral valve stenosis
- Prosthetic heart valve (annuloplasty with or without prosthetic ring, commissurotomy and/or valvuloplasty are permitted)
- Planned cardioversion (electrical or pharmacological)
- Transient atrial fibrillation caused by a reversible disorder (e.g., thyrotoxicosis, PE, recent surgery, MI)
- Known presence of atrial myxoma or left ventricular thrombus
- Active endocarditis

Hemorrhage Risk-Related Criteria

- Active internal bleeding
- History of or condition associated with increased bleeding risk including, but not limited to:
  - Major surgical procedure or trauma within 30 days before the randomization visit
  - Clinically significant gastrointestinal bleeding within 6 months before the randomization visit
  - History of intracranial, intraocular, spinal, or atraumatic intra-articular bleeding
  - Chronic hemorrhagic disorder
  - Known intracranial neoplasm, arteriovenous malformation, or aneurysm
- Planned invasive procedure with potential for uncontrolled bleeding, including major surgery
- Platelet count <90,000/μL at the screening visit
- Sustained uncontrolled hypertension: systolic blood pressure ≥180 mmHg or diastolic blood pressure ≥100 mmHg

Concomitant Conditions and Therapies

- Severe, disabling stroke (modified Rankin score of 4 to 5, inclusive) within 3 months or any stroke within 14 days before the randomization visit
- Transient ischemic attack within 3 days before the randomization visit
- Indication for anticoagulant therapy for a condition other than atrial fibrillation (e.g., VTE)
- Treatment with:
  - Aspirin >100 mg daily
  - Aspirin in combination with thienopyridines within 5 days before randomization
  - Intravenous antiplatelets within 5 days before randomization
  - Fibrinolytics within 10 days before randomization
  - Note: Aspirin ≤100 mg monotherapy is allowed and thienopyridine monotherapy is allowed.
- Anticipated need for chronic treatment with a non-steroidal anti-inflammatory drug
- Systemic treatment with a strong inhibitor of cytochrome P450 3A4, such as ketoconazole or protease inhibitors, within 4 days before randomization, or planned treatment during the time period of the study
- Treatment with a strong inducer of cytochrome P450 3A4, such as rifampin/rifampicin, within 4 days before randomization, or planned treatment during the time period of the study
- Anemia (hemoglobin <10 g/dL) at the screening visit
- Pregnancy or breast-feeding
- Any other contraindication to warfarin
- Known HIV infection at time of screening
- Calculated CLCR <30 mL/min at the screening visit (refer to Attachment 4 for calculating CLCR)
- Known significant liver disease (e.g., acute clinical hepatitis, chronic active hepatitis, cirrhosis), or ALT >3 x the ULN

Study Participation and Follow-up-Related Criteria

- Serious concomitant illness associated with a life expectancy of less than 2 years
- Drug addiction or alcohol abuse within 3 years before the randomization visit

- Have received an experimental drug or used an experimental medical device within 30 days before the planned start of treatment
- Previous randomization in the present study or other study of rivaroxaban
- Known allergy or hypersensitivity to any component of rivaroxaban, warfarin or placebo excipients (includes lactose, microcrystalline cellulose, magnesium stearate, hypromellose, macrogol, croscarmellose sodium, sodium lauryl sulfate, titanium oxide/ferric oxide red, titanium dioxide/ferric oxide red, anhydrous lactose, pregelatinized starch, FD&C Red #6 barium lake, FD&C Yellow #10 aluminum lake, FD&C Blue #1 aluminum lake, FD&C Yellow #6 aluminum lake, cornstarch, lactose monohydrate)
- Inability or unwillingness to comply with study-related procedures
- Employees of the investigator or study center, with direct involvement in the proposed study or other studies under the direction of that investigator or study center, as well as family members of the employees or the investigator.

Follow-up Procedures

The protocol required that all randomized patients be seen at 1, 2, and 4 weeks and monthly thereafter for the duration of the study for measurement of international normalized ratio (INR), surveillance for primary endpoint events, transient ischemic attack, myocardial infarction, medical or surgical procedures, adverse events, and vital status. A standardized questionnaire and examination were used to screen for stroke symptoms and potential clinical events during follow-up. Concomitant use of aspirin up to 100 mg daily was permitted. Treatment with thienopyridine antiplatelet agents was prohibited for 5 days before randomization and throughout the treatment period, except for patients undergoing cardiovascular interventions who were eligible to receive appropriate dual antiplatelet therapy with aspirin and a thienopyridine concomitantly with the assigned anticoagulant at the investigator's discretion.

Outcome Definitions

Stroke was defined as a sudden focal neurologic deficit of presumed cerebrovascular etiology that persisted beyond 24 hours and was not due to another identifiable cause. An event matching this definition but lasting less than 24

hours was considered to be a transient ischemic attack. Brain imaging (computed tomography or magnetic resonance imaging) was recommended for all suspected strokes, and this was performed in 82.1% of patients with strokes.

Systemic embolism was defined as abrupt vascular insufficiency associated with clinical or radiological evidence of arterial occlusion in the absence of another likely mechanism (e.g., atherosclerosis, instrumentation, or trauma).

Myocardial infarction was defined by typical symptoms and cardiac biomarker elevation (troponin I or T, creatine kinase-MB) above the upper limit of normal, new pathological Q waves in at least 2 contiguous electrocardiogram leads, or confirmation at autopsy. Deaths were adjudicated as either vascular (e.g., due to stroke, embolism, myocardial infarction, or arrhythmia) or non-vascular (e.g., malignancy or infection).

Major bleeding was defined as clinically overt bleeding associated with any of the following: fatal outcome, involvement of a critical anatomic site (intracranial, spinal, ocular, pericardial, articular, retroperitoneal, or intramuscular with compartment syndrome), fall in hemoglobin concentration $\geq$2 g/dL, transfusion of $\geq$2 units of whole blood or packed red blood cells, or permanent disability.

Non-major clinically relevant bleeding was defined as overt bleeding not meeting criteria for major bleeding but requiring medical intervention, unscheduled contact (visit or telephone) with a physician, temporary interruption of study drug (i.e., delayed dosing), pain, or impairment of daily activities.

Results

Site Quality Issues

After database lock, the sponsor and executive committee were made aware of potential quality issues at one investigator site that had randomized 20 patients. The site used duplicate copies of ECG tracings as data for more than one subject. The site did undergo an audit during the conduct of the trial and had data verification performed. Each subject was confirmed to be unique. As this information was obtained after the database was locked and the overall study results unblinded, a sensitivity analysis excluding data from this site was performed and demonstrated no material change in the efficacy or safety analyses or conclusions. The decision was made not to alter the primary analysis datasets.

Time to Events in Patients Off Treatment

Of patients ending treatment before the end of the prescribed period (premature permanent discontinuation), 51.2% in the rivaroxaban arm and 48.8% in the warfarin arm subsequently received (open-label) warfarin, and the median time to the first therapeutic INR value was 13 days after cessation of rivaroxaban versus 3 days for those originally assigned to warfarin. During the 30 days following treatment transition, 42 primary events occurred in the rivaroxaban group versus 36 in the warfarin group. Times to first primary event on assigned treatment and after premature discontinuation of treatment are shown in **Figure 2**.

After sites were notified to end study treatment, 92.2% of patients in both groups still on the assigned study drug were transitioned to VKAs, during which the median times to reach therapeutic INR were13 days for those previously assigned to rivaroxaban versus 3 days for the warfarin group. Significantly more patients transitioning from rivaroxaban than from warfarin developed primary events during the first month after termination of randomized treatment (22 vs. 7; P=0.008) (**online Figure 2**).

Non-hemorrhagic Adverse Events

Overall, the rates of treatment-emergent adverse events were similar in both treatment groups, except for epistaxis, which occurred more frequently in the patients treated with rivaroxaban group (**online Table 3).** Elevations in serum levels of hepatic enzymes occurred with equal frequencies in both groups (ALT >3 ×the upper limit of normal [ULN] and TBL >2 × ULN either on same day or within following 30 days in 0.47 vs. 0.50 of patients; HR 0.96; 95 CI, 0.59–1.54).

Online Table 1. Baseline Patient Characteristics: Medications*

Characteristic	**Rivaroxaban (N=7111)**	**Warfarin (N=7125)**
Medications, no. (%)†		
Beta-blockers	4631 (65.12)	4686 (65.77)
Diuretics	4289 (60.32)	4248 (59.62)
Angiotensin-converting enzyme inhibitors	3915 (55.06)	3845 (53.96)
Statins	3055 (42.96)	3077 (43.19)
Digitalis glycosides	2758 (38.78)	2768 (38.85)
Aspirin	2726 (38.33)	2759 (38.72)

*No statistically significant differences to report.
†Medications taken prior to first dose of study drug. Rates are based on the safety on-treatment population.

Online Table 2. Major bleeding by site*

	Rivaroxaban (N=7111)	**Warfarin (N=7125)**
Major bleeding, no. (%)	395 (5.55)	386 (5.42)
Gastrointestinal (upper, lower, and rectal)†	224 (3.15)	154 (2.16)
Intracranial‡	55 (0.77)	84 (1.18)
Intraparenchymal‡	37 (0.52)	56 (0.79)
Non-traumatic‡	33 (0.46)	54 (1.76)
Traumatic	4 (0.06)	2 (0.03)
Intraventricular	2 (0.03)	4 (0.06)
Subdural hematoma	12 (0.17)	22 (0.31)
Subarachnoid	4 (0.06)	1 (0.01)
Epidural hematoma	0	1 (0.01)
Macroscopic hematuria	26 (0.37)	21 (0.29)
Bleeding associated with non-cardiac surgery	19 (0.27)	26 (0.36)
Intraocular/Retinal	17 (0.24)	24 (0.34)
Intraarticular	16 (0.23)	21 (0.29)
Epistaxis	13 (0.18)	14 (0.20)

*Site based on blinded adjudication.
†Combined gastrointestinal bleed rate P<0.001.
‡P<0.05

Online Table 3. Adverse Events*

Adverse Event, no. (%)	**Rivaroxaban (N=7111)**	**Warfarin (N=7125)**
Total patients with treatment-emergent adverse events†	5791 (81.44)	5810 (81.54)
Epistaxis‡	721 (10.14)	609 (8.55)
Peripheral edema	435 (6.12)	444 (6.23)
Dizziness	433 (6.09)	449 (6.30)
Nasopharyngitis	421 (5.92)	455 (6.39)
Cardiac failure	397 (5.58)	420 (5.89)
Bronchitis	396 (5.57)	417 (5.85)
Dyspnea	380 (5.34)	394 (5.53)
Diarrhea	379 (5.33)	397 (5.57)
Cough	343 (4.82)	353 (4.95)
Back pain	338 (4.75)	347 (4.87)
Upper respiratory tract infection	336 (4.73)	325 (4.56)
Headache	324 (4.56)	363 (5.09)
Arthralgia	301 (4.23)	331 (4.65)
Hematuria‡	296 (4.16)	242 (3.40)
Urinary tract infection	293 (4.12)	321 (4.51)
ALT >3 × ULN and TBL >2 × ULN either on same day or within following 30 days	33 (0.47)	35 (0.50)

Based on safety population (B) with N=7111 in the rivaroxaban arm and N=7125 in the warfarin arm.
ALT denotes alanine aminotransferase; TBL, total bilirubin; ULN, upper limit of normal.
*15 most frequent treatment-emergent adverse events based on rivaroxaban treatment group.
†Events that started on or after the first dose of study medication and up to 2 days after the last dose of study medication.
‡P<0.05.

Online Table 4. Secondary Efficacy Outcomes* by Treatment Group†

	Rivaroxaban		Warfarin		Rivaroxaban vs. Warfarin	
	Total	**Event Rate (100 Pt-Yr)‡**	**Total**	**Event Rate (100 Pt-Yr)‡**	**Hazard Ratio (95% CI)§**	**P Value‖**
Secondary efficacy endpoints, no. (%)¶						
Stroke, non-CNS embolism, and vascular death	346 (4.90)	3.11	410 (5.79)	3.63	0.86 (0.74,0.99)	0.034
Stroke, non-CNS embolism, vascular death, and myocardial infarction	433 (6.13)	3.91	519 (7.33)	4.62	0.85 (0.74,0.96)	0.010
Stroke	184 (2.61)	1.65	221 (3.12)	1.96	0.85 (0.70, 1.03)	0.092
Hemorrhagic	29 (0.41)	0.26	50 (0.71)	0.44	0.59 (0.37, 0.93)	0.024
Ischemic	149 (2.11)	1.34	161 (2.27)	1.42	0.94 (0.75, 1.17)	0.581
Unknown	7 (0.10)	0.06	11 (0.16)	0.10	0.65 (0.25, 1.67)	0.366
Stroke outcome#						
Death	47 (0.67)	0.42	67 (0.95)	0.59	0.71 (0.49, 1.03)	0.075
Disabling	43 (0.61)	0.39	57 (0.80)	0.50	0.77 (0.52, 1.14)	0.188
Nondisabling	88 (1.25)	0.79	87 (1.23)	0.77	1.03 (0.76, 1.38)	0.863
Unknown	7 (0.10)	0.06	12 (0.17)	0.11	0.59 (0.23, 1.50)	0.271
Non-CNS systemic embolism, no. (%)	5 (0.07)	0.04	22 (0.31)	0.19	0.23 (0.09, 0.61)	0.003
Myocardial infarction, no. (%)	101 (1.43)	0.91	126 (1.78)	1.12	0.81 (0.63, 1.06)	0.121
All-cause mortality, no. (%)**	208 (2.95)	1.87	250 (3.53)	2.21	0.85 (0.70, 1.02)	0.073
Vascular death	170 (2.41)	1.53	193 (2.73)	1.71	0.89 (0.73, 1.10)	0.289
Non-vascular death	21 (0.30)	0.19	34 (0.48)	0.30	0.63 (0.36, 1.08)	0.094
Unknown	17 (0.24)	0.15	23 (0.32)	0.20	0.75 (0.40, 1.41)	0.370

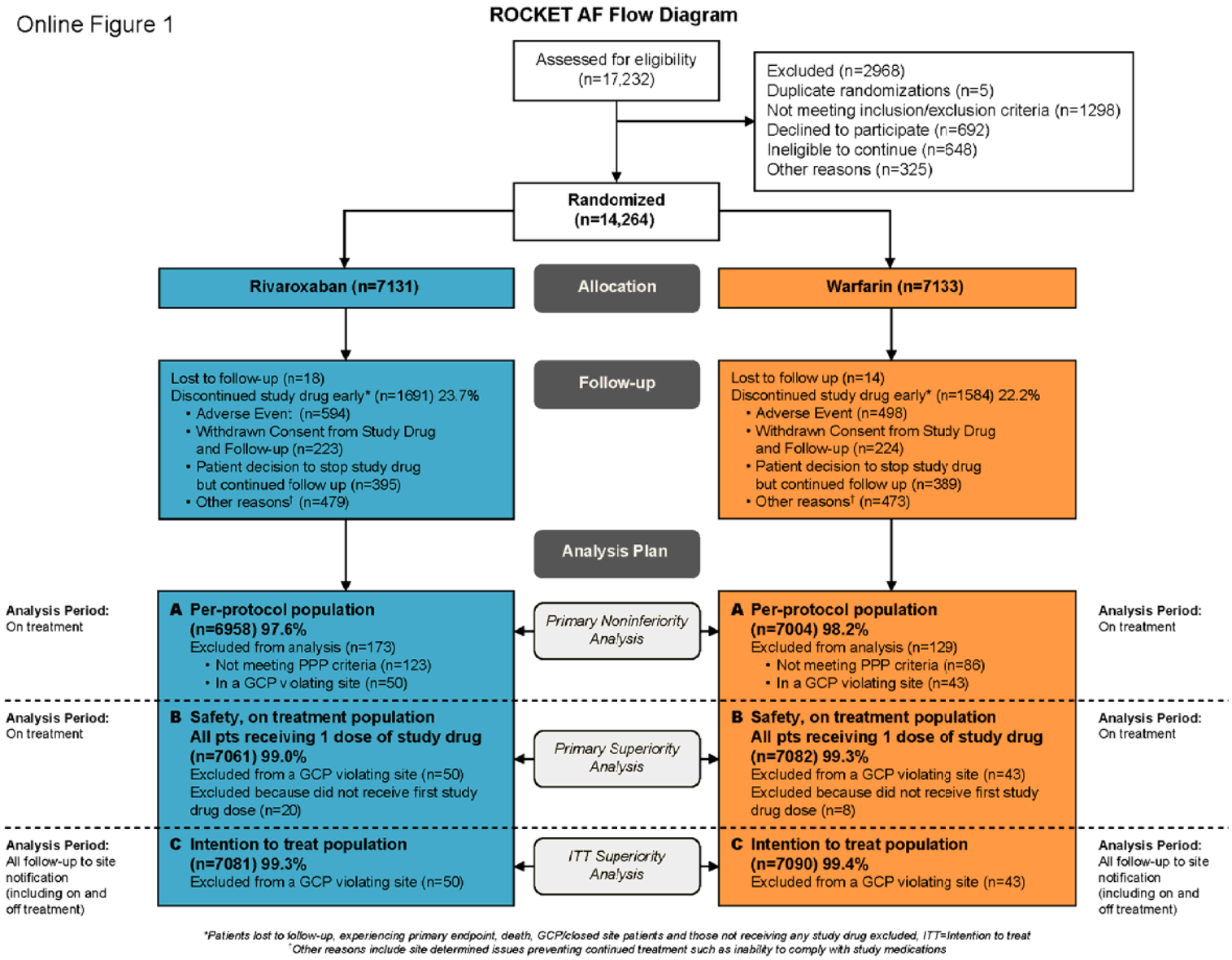
CI denotes confidence interval; Pt, patient; Yr, year.
*Major secondary efficacy endpoint 1 is the composite of stroke, non-CNS systemic embolism, and vascular death. Major secondary efficacy endpoint 2 is the composite of stroke, non-CNS systemic embolism, myocardial infarction, and vascular death.
†Based on safety on-treatment population (B). One GCP violating site was not used for efficacy analyses resulting in N=7061 in the rivaroxaban arm and N=7082 in the warfarin arm.
‡Number of events per 100 patient-years of follow-up.
§Hazard ratio (95% CI) and P value from Cox proportional hazard model with treatment group as a covariate.
‖P value (2-sided) for superiority of rivaroxaban versus warfarin in hazard ratio.
¶Safety on-treatment population.
#Stroke outcome is based on investigator's assessment of modified Rankin scale score: 0–2=nondisabling, 3–5=disabling, 6=death.
**All-cause mortality data are presented based on the Safety population (excluding GCP violating site).

Online Table 5. Treatment Effects by Quartiles of Center Time in Therapeutic Range*†

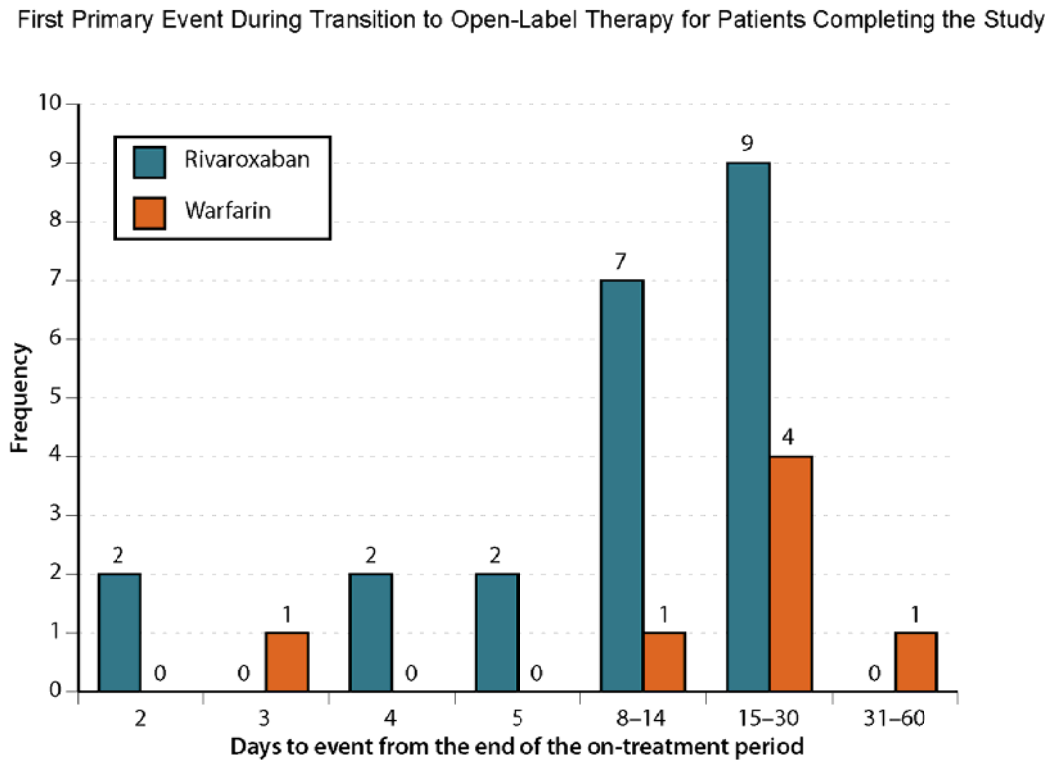
	Rivaroxaban		Warfarin		Rivaroxaban vs. Warfarin
Center TTR‡	**Total**	**Event Rate (100 Pt-Yr)§**	**Total**	**Event Rate (100 Pt-Yr)§**	**Hazard Ratio (95% CI)‖**
0.00–50.6%	45/1735 (2.59)	1.77	62/1689 (3.67)	2.53	0.70 (0.48, 1.03)
50.7–58.5%	53/1746 (3.04)	1.94	63/1807 (3.49)	2.18	0.89 (0.62, 1.29)
58.6–65.7%	54/1734 (3.11)	1.90	62/1758 (3.53)	2.14	0.89 (0.62, 1.28)
65.7–100.0%	37/1676 (2.21)	1.33	55/1826 (3.01)	1.80	0.74 (0.49, 1.12)

Values presented as no. of subjects with events/no. of subjects in each subgroup (%). Safety population: N=7061 for the rivaroxaban arm and N=7082 for the warfarin arm.
CI denotes confidence interval; Pt, patient; TTR, time in therapeutic range; Yr, year.
*P value for interaction=0.736.
†Time in therapeutic range=2–3 inclusive.
‡Center TTR is calculated using total number of international normalized ratio (INR) values in target range from all warfarin subjects within a center divided by total number of INR values from all warfarin subjects within the center.
§Number of events per 100 patient-years of follow-up.
‖Hazard ratio (95% CI) from the Cox proportional hazard model with treatment as a covariate.

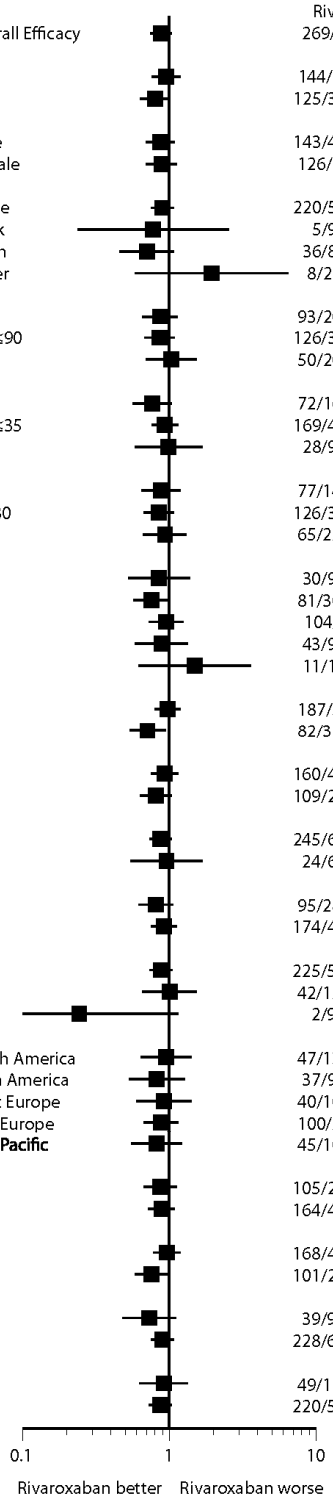
Online Figure 1



Online Figure 2



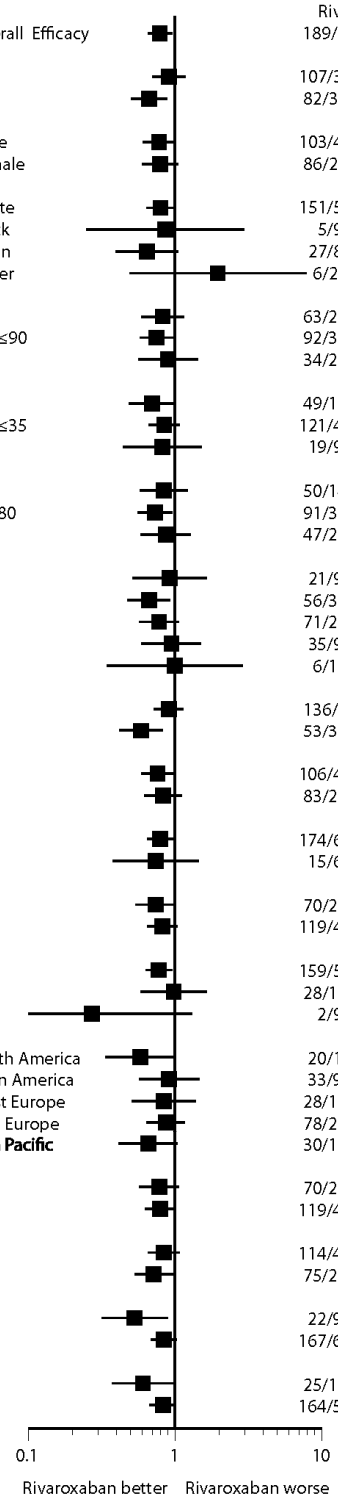
Online Figure 3
ITT to Site Notification



		Rivaroxaban	Warfarin	Hazard Ratio	P-value*
	Overall Efficacy	269/7081 (3.8%)	306/7090 (4.32%)	0.88 (0.75, 1.03)	
Age	<75	144/3999 (3.6%)	152/4008 (3.79%)	0.95 (0.76, 1.19)	0.313
	≥75	125/3082 (4.06%)	154/3082 (5%)	0.8 (0.63, 1.02)	
Sex	Male	143/4279 (3.34%)	164/4287 (3.83%)	0.87 (0.7, 1.09)	0.927
	Female	126/2802 (4.5%)	142/2803 (5.07%)	0.89 (0.7, 1.12)	
Race	White	220/5872 (3.75%)	246/5914 (4.16%)	0.9 (0.75, 1.08)	0.424
	Black	5/94 (5.32%)	6/86 (6.98%)	0.78 (0.24, 2.55)	
	Asian	36/897 (4.01%)	50/889 (5.62%)	0.7 (0.46, 1.08)	
	Other	8/218 (3.67%)	4/201 (1.99%)	1.95 (0.59, 6.49)	
Weight (kg)	≤70	93/2013 (4.62%)	109/2012 (5.42%)	0.86 (0.65, 1.14)	0.713
	70–≤90	126/3031 (4.16%)	151/3135 (4.82%)	0.86 (0.68, 1.08)	
	>90	50/2035 (2.46%)	46/1942 (2.37%)	1.03 (0.69, 1.54)	
BMI (kg/m^2)	≤25	72/1695 (4.25%)	98/1750 (5.6%)	0.77 (0.56, 1.04)	0.537
	25–≤35	169/4409 (3.83%)	180/4417 (4.08%)	0.93 (0.76, 1.15)	
	>35	28/972 (2.88%)	27/919 (2.94%)	0.99 (0.58, 1.68)	
CrCl (mL/min)	<50	77/1490 (5.17%)	86/1459 (5.89%)	0.88 (0.65, 1.19)	0.900
	50–80	126/3298 (3.82%)	151/3400 (4.44%)	0.85 (0.67, 1.08)	
	>80	65/2285 (2.84%)	68/2222 (3.06%)	0.94 (0.67, 1.31)	
CHADS2	2	30/924 (3.25%)	36/933 (3.86%)	0.85 (0.52, 1.38)	0.603
	3	81/3036 (2.67%)	109/3133 (3.48%)	0.76 (0.57, 1.01)	
	4	104/2078 (5%)	105/1989 (5.28%)	0.95 (0.72, 1.24)	
	5	43/920 (4.67%)	47/877 (5.36%)	0.88 (0.58, 1.34)	
	6	11/122 (9.02%)	9/156 (5.77%)	1.49 (0.62, 3.59)	
Prior Stroke/TIA/ Non-CNS Systemic Embolism	Yes	187/3892 (4.8%)	190/3875 (4.9%)	0.98 (0.8, 1.2)	0.072
	No	82/3189 (2.57%)	116/3215 (3.61%)	0.71 (0.54, 0.94)	
Congestive Heart Failure	Yes	160/4438 (3.61%)	172/4413 (3.9%)	0.93 (0.75, 1.15)	0.419
	No	109/2642 (4.13%)	134/2676 (5.01%)	0.81 (0.63, 1.04)	
Hypertension	Yes	245/6389 (3.83%)	282/6435 (4.38%)	0.87 (0.73, 1.03)	0.761
	No	24/692 (3.47%)	24/655 (3.66%)	0.96 (0.54, 1.69)	
Diabetes	Yes	95/2851 (3.33%)	114/2796 (4.08%)	0.81 (0.62, 1.07)	0.483
	No	174/4230 (4.11%)	192/4294 (4.47%)	0.92 (0.75, 1.13)	
AF Type	Persistent	225/5754 (3.91%)	255/5731 (4.45%)	0.88 (0.73, 1.05)	0.218
	Paroxysmal	42/1231 (3.41%)	43/1259 (3.42%)	1 (0.66, 1.54)	
	Newly Diagnosed	2/96 (2.08%)	8/100 (8%)	0.24 (0.05, 1.14)	
Region	North America	47/1339 (3.51%)	50/1342 (3.73%)	0.95 (0.64, 1.42)	0.982
	Latin America	37/940 (3.94%)	45/938 (4.8%)	0.82 (0.53, 1.27)	
	West Europe	40/1046 (3.82%)	43/1050 (4.1%)	0.92 (0.6, 1.41)	
	East Europe	100/2701 (3.7%)	114/2706 (4.21%)	0.88 (0.67, 1.15)	
	Asia Pacific	45/1055 (4.27%)	54/1054 (5.12%)	0.82 (0.55, 1.22)	
Prior ASA Use	Yes	105/2575 (4.08%)	121/2609 (4.64%)	0.87 (0.67, 1.13)	0.905
	No	164/4506 (3.64%)	185/4481 (4.13%)	0.88 (0.72, 1.09)	
Prior VKA Use	Yes	168/4413 (3.81%)	175/4440 (3.94%)	0.97 (0.78, 1.19)	0.160
	No	101/2668 (3.79%)	131/2650 (4.94%)	0.76 (0.59, 0.98)	
Prior PPI Use	Yes	39/909 (4.29%)	50/882 (5.67%)	0.73 (0.48, 1.11)	0.390
	No	228/6152 (3.71%)	256/6200 (4.13%)	0.9 (0.75, 1.07)	
Prior MI	Yes	49/1173 (4.18%)	58/1273 (4.56%)	0.92 (0.63, 1.34)	0.805
	No	220/5908 (3.72%)	248/5817 (4.26%)	0.87 (0.73, 1.04)	

*P-value for interaction

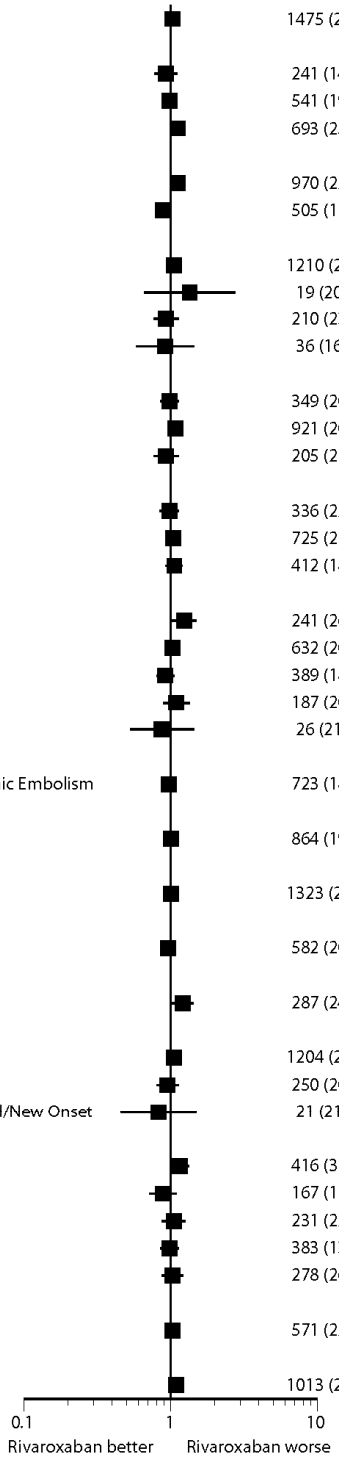
Online Figure 4
Safety on Treatment



		Rivaroxaban	Warfarin	Hazard Ratio	P-value*
	Overall Efficacy	189/7061 (2.68%)	243/7082 (3.43%)	0.79 (0.65, 0.96)	
Age	<75	107/3988 (2.68%)	119/4005 (2.97%)	0.91 (0.7, 1.19)	0.107
	≥75	82/3073 (2.67%)	124/3077 (4.03%)	0.67 (0.51, 0.88)	
Sex	Male	103/4270 (2.41%)	136/4283 (3.18%)	0.78 (0.6, 1.01)	0.922
	Female	86/2791 (3.08%)	107/2799 (3.82%)	0.8 (0.6, 1.06)	
Race	White	151/5856 (2.58%)	194/5909 (3.28%)	0.8 (0.64, 0.99)	0.486
	Black	5/94 (5.32%)	5/85 (5.88%)	0.86 (0.25, 2.97)	
	Asian	27/894 (3.02%)	41/887 (4.62%)	0.64 (0.4, 1.05)	
	Other	6/217 (2.76%)	3/201 (1.49%)	1.96 (0.49, 7.84)	
Weight (kg)	≤70	63/2004 (3.14%)	78/2008 (3.88%)	0.83 (0.59, 1.15)	0.778
	70–≤90	92/3022 (3.04%)	129/3133 (4.12%)	0.75 (0.58, 0.98)	
	>90	34/2033 (1.67%)	36/1940 (1.86%)	0.9 (0.56, 1.44)	
BMI (kg/m^2)	≤25	49/1685 (2.91%)	75/1745 (4.3%)	0.7 (0.49, 1)	0.692
	25–≤35	121/4400 (2.75%)	145/4415 (3.28%)	0.84 (0.66, 1.07)	
	>35	19/971 (1.96%)	22/918 (2.4%)	0.82 (0.45, 1.52)	
CrCl (mL/min)	<50	50/1485 (3.37%)	60/1456 (4.12%)	0.84 (0.58, 1.23)	0.715
	50–80	91/3290 (2.77%)	128/3396 (3.77%)	0.73 (0.56, 0.96)	
	>80	47/2278 (2.06%)	54/2221 (2.43%)	0.87 (0.59, 1.28)	
CHADS2	2	21/922 (2.28%)	24/931 (2.58%)	0.92 (0.51, 1.66)	0.739
	3	56/3025 (1.85%)	87/3131 (2.78%)	0.67 (0.48, 0.93)	
	4	71/2073 (3.42%)	88/1988 (4.43%)	0.78 (0.57, 1.07)	
	5	35/918 (3.81%)	36/875 (4.11%)	0.95 (0.59, 1.51)	
	6	6/122 (4.92%)	8/155 (5.16%)	1 (0.35, 2.88)	
Prior Stroke/TIA/ Non-CNS Systemic Embolism	Yes	136/3881 (3.5%)	151/3869 (3.9%)	0.91 (0.72, 1.14)	0.039
	No	53/3180 (1.67%)	92/3213 (2.86%)	0.59 (0.42, 0.83)	
Congestive Heart Failure	Yes	106/4428 (2.39%)	141/4409 (3.2%)	0.76 (0.59, 0.98)	0.664
	No	83/2632 (3.15%)	102/2672 (3.82%)	0.83 (0.62, 1.11)	
Hypertension	Yes	174/6372 (2.73%)	223/6429 (3.47%)	0.79 (0.65, 0.97)	0.850
	No	15/689 (2.18%)	20/653 (3.06%)	0.74 (0.38, 1.45)	
Diabetes	Yes	70/2842 (2.46%)	94/2793 (3.37%)	0.74 (0.54, 1.01)	0.597
	No	119/4219 (2.82%)	149/4289 (3.47%)	0.82 (0.65, 1.05)	
AF Type	Persistent	159/5739 (2.77%)	206/5723 (3.6%)	0.78 (0.63, 0.96)	0.300
	Paroxysmal	28/1228 (2.28%)	30/1259 (2.38%)	0.98 (0.59, 1.64)	
	Newly Diagnosed	2/94 (2.13%)	7/100 (7%)	0.27 (0.06, 1.32)	
Region	North America	20/1334 (1.5%)	36/1339 (2.69%)	0.58 (0.34, 1.01)	0.618
	Latin America	33/939 (3.51%)	37/938 (3.94%)	0.91 (0.57, 1.46)	
	West Europe	28/1040 (2.69%)	34/1049 (3.24%)	0.84 (0.51, 1.39)	
	East Europe	78/2696 (2.89%)	91/2704 (3.37%)	0.87 (0.64, 1.17)	
	Asia Pacific	30/1052 (2.85%)	45/1052 (4.28%)	0.66 (0.41, 1.04)	
Prior ASA Use	Yes	70/2567 (2.73%)	91/2606 (3.49%)	0.78 (0.57, 1.07)	0.941
	No	119/4494 (2.65%)	152/4476 (3.4%)	0.79 (0.62, 1.01)	
Prior VKA Use	Yes	114/4401 (2.59%)	140/4437 (3.16%)	0.84 (0.66, 1.08)	0.420
	No	75/2660 (2.82%)	103/2645 (3.89%)	0.72 (0.53, 0.97)	
Prior PPI Use	Yes	22/909 (2.42%)	40/882 (4.54%)	0.53 (0.32, 0.89)	0.113
	No	167/6152 (2.71%)	203/6200 (3.27%)	0.84 (0.69, 1.03)	
Prior MI	Yes	25/1169 (2.14%)	46/1269 (3.62%)	0.61 (0.37, 0.99)	0.252
	No	164/5892 (2.78%)	197/5813 (3.39%)	0.83 (0.67, 1.02)	

*P-value for interaction

Online Figure 5



Major and Non-Major Clinically Relevant Bleeding While on Treatment		Rivaroxaban	Warfarin	HR (95% CI)	P-value*
Overall		1475 (20.74)	1449 (20.34)	1.03 (0.96, 1.11)	
Age	<65	241 (14.64)	260 (15.83)	0.93 (0.78, 1.11)	0.118
	65 to 75	541 (19.48)	556 (19.99)	0.98 (0.87, 1.10)	
	>75	693 (25.78)	633 (23.43)	1.12 (1.00, 1.25)	
Sex	Male	970 (22.60)	898 (20.89)	1.12 (1.02, 1.22)	0.004
	Female	505 (17.91)	551 (19.50)	0.89 (0.79, 1.01)	
Race	White	1210 (20.49)	1178 (19.79)	1.05 (0.97, 1.13)	0.591
	Black	19 (20.21)	13 (15.29)	1.35 (0.66, 2.75)	
	Asian	210 (23.49)	220 (24.80)	0.93 (0.77, 1.13)	
	Other	36 (16.59)	38 (18.91)	0.92 (0.58, 1.44)	
BMI (kg/m^2)	≤25	349 (20.54)	374 (21.26)	0.98 (0.85, 1.14)	0.310
	25 to ≤35	921 (20.78)	865 (19.47)	1.08 (0.98, 1.18)	
	>35	205 (21.03)	209 (22.72)	0.93 (0.77, 1.13)	
CrCl (mL/min)	<50	336 (22.37)	342 (23.17)	0.98 (0.84, 1.14)	0.735
	50 to 80	725 (21.88)	719 (21.09)	1.04 (0.93, 1.15)	
	>80	412 (18.01)	388 (17.40)	1.06 (0.92, 1.21)	
CHADS2	2	241 (26.11)	208 (22.32)	1.24 (1.03, 1.50)	0.121
	3	632 (20.74)	636 (20.15)	1.03 (0.92, 1.15)	
	4	389 (18.64)	402 (20.12)	0.92 (0.80, 1.06)	
	5	187 (20.11)	165 (18.77)	1.09 (0.89, 1.35)	
	6	26 (21.14)	38 (24.05)	0.87 (0.53, 1.44)	
Prior Stroke/TIA/Non-CNS Systemic Embolism		723 (18.51)	739 (19.00)	0.97 (0.88, 1.08)	0.118
Congestive Heart Failure		864 (19.39)	859 (19.36)	1.01 (0.92, 1.11)	0.587
Hypertension		1323 (20.61)	1322 (20.44)	1.01 (0.94, 1.09)	0.183
Diabetes		582 (20.29)	596 (21.18)	0.96 (0.86, 1.08)	0.144
Prior Myocardial Infarction		287 (24.36)	268 (20.90)	1.21 (1.02, 1.43)	0.035
AF Type	Persistent	1204 (20.86)	1153 (20.04)	1.05 (0.97, 1.14)	0.406
	Paroxysmal	250 (20.13)	272 (21.43)	0.95 (0.80, 1.13)	
	Newly Diagnosed/New Onset	21 (21.43)	24 (23.53)	0.83 (0.46, 1.49)	
Region	North America	416 (31.18)	382 (28.53)	1.16 (1.01, 1.33)	0.277
	Latin America	167 (17.78)	185 (19.72)	0.89 (0.72, 1.10)	
	Western Europe	231 (22.21)	225 (21.45)	1.05 (0.87, 1.26)	
	Eastern Europe	383 (13.95)	387 (14.09)	0.98 (0.85, 1.13)	
	Asia Pacific	278 (26.43)	270 (25.67)	1.03 (0.87, 1.22)	
Prior ASA Use		571 (22.15)	558 (21.33)	1.03 (0.92, 1.16)	0.934
Prior VKA Use		1013 (22.86)	965 (21.65)	1.09 (0.99, 1.19)	0.044

*P-value for interaction